

The Gram Point Hierarchy and the Lambert W Generator

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Abstract

The nontrivial zeros of the Riemann zeta function and their relationship to the Gram points are governed by a hierarchy of phase offset functions $\psi_k(t) = \vartheta(t) - kt$, where ϑ is the Riemann–Siegel theta function. We derive explicit closed-form expressions for the level- k Gram sequence $\{g_n^{(k)}\}$ in terms of the Lambert W function and show that all correction terms are generated by iterating a single base inversion operator Ψ_0 composed with the Stirling residual tail of ϑ . The full Lagrange–Bürmann inversion produces an exact power series for $g_n^{(k)}$ in which every coefficient is a rational function of $W((n + \frac{1}{8})e^{-(2k+1)})$, generated by a three-term recurrence on polynomials q_j . The k -th level process $X_k(n) = n\pi - g_n^{(k-1)}$ is thereby expressed entirely in closed form through the same Lambert W generator at shifted level.

1 Definitions and Setup

Definition 1.1. The Riemann–Siegel theta function is

$$\vartheta(t) = \arg\left(\pi^{-it/2} \Gamma\left(\frac{1}{4} + \frac{it}{2}\right)\right), \quad (1)$$

normalized so that $\vartheta(t)$ is real and continuous for $t > 0$ with $\vartheta(0) = 0$.

Definition 1.2. For each integer $k \geq 0$, the k -th phase offset function is

$$\psi_k(t) = \vartheta(t) - kt. \quad (2)$$

Definition 1.3. The Stirling residual tail of ϑ is the function $\varphi: (0, \infty) \rightarrow \mathbb{R}$ defined by the exact identity

$$\varphi(t) = \operatorname{Im}\left[\log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right)\right] - \frac{t}{2} \log \frac{t}{2\pi} + \frac{t}{2} + \frac{\pi}{8}, \quad (3)$$

so that

$$\vartheta(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \varphi(t). \quad (4)$$

The function φ is smooth on $(0, \infty)$ and has the asymptotic expansion

$$\varphi(t) \sim \sum_{j=1}^{\infty} \frac{(-1)^{j+1} B_{2j}}{2j(2j-1)} \left(\frac{2}{t}\right)^{2j-1} = \frac{1}{48t} + \frac{7}{5760t^3} + \frac{31}{80640t^5} + \cdots \quad (5)$$

where B_{2j} are Bernoulli numbers.

Definition 1.4. For each $k \geq 0$, the level- k Gram sequence $\{g_n^{(k)}\}_{n \geq 0}$ is the sequence of positive real numbers satisfying

$$\psi_k(g_n^{(k)}) = n\pi, \quad n = 0, 1, 2, \dots \quad (6)$$

For $k = 0$ this gives the ordinary Gram points $g_n^{(0)} = g_n$ satisfying $\vartheta(g_n) = n\pi$.

Definition 1.5. The k -th level process is

$$X_k(n) = \psi_k(g_n^{(k-1)}) = \vartheta(g_n^{(k-1)}) - k g_n^{(k-1)}, \quad k \geq 1. \quad (7)$$

Definition 1.6. The base inversion function is

$$\Psi_0(c, k) = \frac{2c}{W\left(\frac{c}{\pi} e^{-(2k+1)}\right)}, \quad (8)$$

where W is the principal branch of the Lambert W function.

2 The Explicit Gram Sequence

Theorem 2.1 (Explicit form of X_k). *For every $k \geq 1$ and every $n \geq 0$:*

$$X_k(n) = n\pi - g_n^{(k-1)}. \quad (9)$$

Proof. From definition (6) at level $k - 1$:

$$\vartheta(g_n^{(k-1)}) - (k-1)g_n^{(k-1)} = n\pi, \quad (10)$$

so $\vartheta(g_n^{(k-1)}) = n\pi + (k-1)g_n^{(k-1)}$. Substituting into (7):

$$X_k(n) = n\pi + (k-1)g_n^{(k-1)} - k g_n^{(k-1)} = n\pi - g_n^{(k-1)}. \quad (11)$$

□

Theorem 2.2 (Leading Lambert W solution). *Set $c = \pi(n + \frac{1}{8})$ and $w_k = W(\frac{c}{\pi} e^{-(2k+1)})$. The leading-order solution to (6) is*

$$g_n^{(k)} = \Psi_0(c, k) + O(n^{-1}) = \frac{2\pi(n + \frac{1}{8})}{w_k} + O(n^{-1}). \quad (12)$$

Proof. Substituting (4) into (6):

$$\frac{t}{2} \log \frac{t}{2\pi} - \frac{(2k+1)}{2} t = n\pi + \frac{\pi}{8} - \varphi(t). \quad (13)$$

Drop $\varphi(t)$ at leading order and set $t = 2\pi e^L$, so that $\log(t/2\pi) = L$:

$$\pi e^L (L - (2k+1)) = c. \quad (14)$$

Set $w = L - (2k + 1)$, so $L = w + 2k + 1$:

$$\pi e^{2k+1} w e^w = c, \quad w e^w = \frac{c}{\pi} e^{-(2k+1)}. \quad (15)$$

Therefore $w = W\left(\frac{c}{\pi} e^{-(2k+1)}\right) = w_k$, giving $L = w_k + 2k + 1$ and

$$t = 2\pi e^{w_k+2k+1} = 2\pi e^{2k+1} e^{w_k}. \quad (16)$$

Applying the identity $e^{W(z)} = z/W(z)$:

$$t = 2\pi e^{2k+1} \cdot \frac{c/(\pi e^{2k+1})}{w_k} = \frac{2c}{w_k}. \quad (17)$$

The $O(n^{-1})$ error arises from dropping $\varphi(t) = O(t^{-1})$ in (13). $\square$

3 The Lambert W Generator for All Corrections

Theorem 3.1 (Derivatives of Ψ_0). *Let $w = W\left(\frac{c}{\pi} e^{-(2k+1)}\right)$. The j -th partial derivative of $\Psi_0(c, k)$ with respect to c is*

$$\partial_c^j \Psi_0(c, k) = \frac{2}{(\pi e^{2k+1})^{j-1}} \cdot \frac{q_j(w)}{(1+w)^{2j-1}}, \quad (18)$$

where the polynomials q_j satisfy the recurrence

$$q_1(w) = 1, \quad q_{j+1}(w) = q'_j(w)(1+w) - (2j-1)q_j(w), \quad (19)$$

with initial values

$$q_1 = 1, \quad q_2 = -1, \quad q_3 = 2 + w, \quad q_4 = -(6 + 8w + 3w^2). \quad (20)$$

Proof. From (8), with $z = \frac{c}{\pi} e^{-(2k+1)}$ so that $\partial_c z = \frac{1}{\pi} e^{-(2k+1)}$ and $w = W(z)$:

$$\partial_c w = \frac{w}{c(1+w)}. \quad (21)$$

For $j = 1$, differentiating $\Psi_0 = 2c/w$:

$$\partial_c \Psi_0 = \frac{2}{w} - \frac{2c}{w^2} \cdot \frac{w}{c(1+w)} = \frac{2}{w} - \frac{2}{w(1+w)} = \frac{2}{1+w}, \quad (22)$$

confirming (18) with $q_1 = 1$. Differentiating $\partial_c^j \Psi_0 = 2(\pi e^{2k+1})^{1-j} q_j(w)(1+w)^{-(2j-1)}$ with respect to c and using (21):

$$\begin{aligned} \partial_c^{j+1} \Psi_0 &= \frac{2}{(\pi e^{2k+1})^{j-1}} \cdot \frac{1}{(1+w)^{2j-1}} \cdot \partial_c [q_j(w)] - \frac{2(2j-1)}{(\pi e^{2k+1})^{j-1}} \cdot \frac{q_j(w)}{(1+w)^{2j}} \cdot \partial_c w \\ &= \frac{2}{(\pi e^{2k+1})^j} \cdot \frac{q'_j(w)(1+w) - (2j-1)q_j(w)}{(1+w)^{2j+1}}. \end{aligned} \quad (23)$$

Defining $q_{j+1}(w) = q'_j(w)(1+w) - (2j-1)q_j(w)$ establishes (19) and (18) by induction. $\square$

Theorem 3.2 (Exact Gram sequence via Lagrange–Bürmann inversion). *Define the correction function*

$$\eta_k(c) = \varphi(\Psi_0(c, k)), \quad (24)$$

where φ is defined by (3). Then the exact solution to (6) is

$$g_n^{(k)} = \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} \frac{d^{j-1}}{dc^{j-1}} [\partial_c \Psi_0(c, k) \cdot \eta_k(c)^j]_{c=\pi(n+1/8)}, \quad (25)$$

where $\frac{d^{-1}}{dc^{-1}}$ at $j = 0$ denotes the identity operator, and all derivatives of Ψ_0 are given by (18).

Proof. Equation (6) with (4) is

$$f(t) \frac{t}{2} \log \frac{t}{2\pi} - \frac{(2k+1)}{2} t = c - \varphi(t), \quad (26)$$

with $c = \pi(n + \frac{1}{8})$. The function $\Psi_0(\cdot, k)$ is the exact inverse of f : $f(\Psi_0(c, k)) = c$ for all c in the range of f . Equation (26) is therefore the fixed-point equation

$$t = \Psi_0(c - \varphi(t), k). \quad (27)$$

Writing $t = \Psi_0(c, k) + \delta$, the Lagrange–Bürmann theorem applied to (27) with expansion variable φ gives the series (25). Convergence follows from the analyticity of Ψ_0 and φ in a neighborhood of c for c away from the branch point of W at $-1/e$, which is excluded for $n \geq 0$. The coefficients of φ^j are expressible via (18) and (19), giving a rational function of w_k at each order. $\square$

Corollary 3.3 (First two explicit correction terms). *Writing $w = w_k = W((n + \frac{1}{8})e^{-(2k+1)})$ and $\Psi = \frac{2\pi(n+1/8)}{w}$:*

$$g_n^{(k)} = \Psi - \frac{2}{1+w} \cdot \varphi(\Psi) + \frac{1}{(\pi e^{2k+1})(1+w)^3} \cdot \varphi(\Psi)^2 + O(\varphi(\Psi)^3), \quad (28)$$

where $\varphi(\Psi)$ is given by the exact expression (3) evaluated at Ψ .

Proof. Evaluate (25) at $j = 0, 1, 2$ using $\partial_c \Psi_0 = 2/(1+w)$ from Theorem 3.1. $\square$

Corollary 3.4 (Explicit k -th level process). *The k -th level process $X_k(n) = n\pi - g_n^{(k-1)}$ is*

$$X_k(n) = n\pi - \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} \frac{d^{j-1}}{dc^{j-1}} \left[\frac{2q_1(w_{k-1})}{1+w_{k-1}} \cdot \eta_{k-1}(c)^j \right]_{c=\pi(n+1/8)}, \quad (29)$$

with $w_{k-1} = W((n + \frac{1}{8})e^{-(2k-1)})$ and $\eta_{k-1}(c) = \varphi(\Psi_0(c, k-1))$. For $k = 1$:

$$X_1(n) = n\pi - \frac{2\pi(n + \frac{1}{8})}{W((n + \frac{1}{8})/e)} + \frac{2}{1+w_0} \cdot \varphi\left(\frac{2\pi(n + \frac{1}{8})}{w_0}\right) + O(\varphi^2), \quad (30)$$

with $w_0 = W((n + \frac{1}{8})/e)$.

Proof. Substitute (25) with k replaced by $k - 1$ into (9). □

Remark 3.5. The polynomials q_j in (19) depend only on w and not on k . The entire dependence on the level k enters through the argument of W : replacing k by $k + 1$ shifts the argument by e^{-2} . Every level of the hierarchy is therefore generated by the same algebraic structure — the recurrence (19) and the base function Ψ_0 — evaluated at a geometrically shifted argument of W .